

M. TAREK HAMWI

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SUMMARY

I'm a Software Engineering student at APU with experience in R&D, e-commerce, and industrial internship work, where I contributed to system design, mobile application migration, and web development. During my internship at Kloudius Services, I migrated React Native TV applications to Amazon's Vega OS platform and built a production-grade UI Kit. I earned 2nd place in the Joget NextGen Hackathon 2024 and am passionate about scalable system design, cross-platform development, and team collaboration.

WORK EXPERIENCE

Scrum Master, Research and Development Team (R&D)

2024 – Present

Centre of Technology & Innovation – APU | Kuala Lumpur, Malaysia

- Serve as Scrum Master for APCentral, a centralized academic workflow system, overseeing project progress and facilitating team collaboration.
- Collaborated with cross-functional teams to ensure the system's scalability and efficiency, resuming and continuing to lead the role following completion of an industrial internship.

Software Engineering Intern (React Native & Amazon Vega OS)

Jan 2026 – Apr 2026

Kloudius Services Sdn. Bhd. | Bangsar South, Kuala Lumpur, Malaysia

- Migrated multiple React Native TV applications, including a fully functional Netflix-clone app with video playback, from an Android-based environment to Amazon's Vega OS platform.
- Designed and built a comprehensive Vega OS UI Kit as a structured React Native codebase, covering design tokens, a 12-column responsive grid, reusable components (Header, Button system, Sidebar Navigation, Focusable Row/List), and Redux-based state management.
- Integrated and debugged three JavaScript media players (Shaka Player, HLS.js, DASH.js) with full D-pad remote support, resolving platform-specific playback crashes and polyfill incompatibilities.
- Built and documented demonstration screens for 30+ Expo and React Native libraries, validated on physical Fire TV Stick and Echo Show 8 hardware.
- Presented project progress in two formal stakeholder demonstrations and produced technical scripts/blog content on migration strategy and performance optimisation for Vega OS.

E-commerce

2022 – 2023

Ck Decor | Malaysia – Kota Bharu

- Built and enhanced the company's online profile on Shopee and Lazada.
- Created targeted advertisements for Facebook and Instagram to boost brand visibility.
- Assisted in developing the company's website, gaining hands-on experience in web design.

EDUCATION

BSc (Hons) in Software Engineering

2023 –

Asia Pacific University (APU) | Malaysia – Bukit Jalil

- Expected Graduation: February 2027
- Current CGPA: 3.56

FOUNDATION

2022 – 2023

MAHSA University | Malaysia – Saujana Putra

- Qualification: Foundation In Science
- CGPA: 3.53

IGCSE – High School

2019 – 2021

Wadi Sofia School | Malaysia – Kota Bharu

- A*: Mathematics, Chemistry, Physics, Arabic
- A: Information and Communication Technology (ICT), Additional Mathematics, Biology

PROJECTS

ParKL – Cross-Platform Smart Parking Management System (Final Year Project)

Using React Native, Flask, PostgreSQL, AI-assisted computer vision

- Designed and developed ParKL, a cross-platform smart parking system comprising a driver-facing mobile app and an administrator web dashboard, addressing excessive parking search time and poor visibility of specialised OKU/EV/women's bays.
- Implemented real-time bay availability on an interactive floor map, preference-based spot recommendation, voice-guided bay reassignment alerts, and a "find my car" location memory feature.
- Built an AI-assisted floor plan onboarding tool that lets administrators automatically extract parking spot coordinates from an uploaded facility image.
- Conducted a 95-respondent user survey to validate the problem and derive requirements; 66% of respondents reported frequently struggling to find parking, directly shaping feature prioritisation.
- Aligned the system with UN Sustainable Development Goal 11 (Sustainable Cities and Communities) by targeting reduced search-related fuel waste and improved accessibility.

Vega OS UI Kit & React Native TV Migration (Internship Project)

Using React Native, TypeScript, Redux, Shaka Player, HLS.js, DASH.js

- Successfully migrated 2 React Native TV applications, a general TV UI sample and a full Netflix-clone app with video playback, to be fully compatible with Amazon's Vega OS platform.
- Researched Amazon's Vega OS platform from first principles and produced a documented migration reference repository for React Native TV applications.
- Built a production-grade UI Kit with design tokens, component architecture, and validated media player integrations, presented to stakeholders in two formal demos.

APCentral

Using Angular, Flask, AWS, Postman, Figma

- Contributed as part of the developer team to build APCentral, a centralized academic workflow system integrating platforms like iSmart, APTIME, GIMS, Moodle, and Exam Scheduling tools.
- Created and handled the complete intake flow and exam flow processes end-to-end within the system.
- Helped implement automation features to reduce manual data entry, minimize human error, and improve tracking of academic processes.
- Supported the digitization of key academic operations such as class scheduling, lecturer allocation, and assessment tracking alongside APU CTI's R&D team.

Concurrent Airport Simulation

Using Java Language, Concurrency Concepts

- Designed and developed a system to simulate airport operations, incorporating concurrency concepts like semaphores and thread synchronization.
- Implemented emergency landing priority, exclusive refueling, and safety checks.

Procurement Order Tracking System (POTS)

Using Java Language

- Developed a desktop application to streamline procurement tracking.
- Implemented real-time status updates, and search filters.
- Helped improve operational efficiency and reduce delays in item processing.

Delicious Restaurant Management System (DRMS)

Using C# Language

- Contributed to the development of a restaurant management system allowing admins, managers, chefs, and customers to manage users, reservations, and orders.
- Implemented object-oriented concepts such as classes, methods, and objects for various user roles.
- Utilized Microsoft SQL Server to store and manage data for menu categories, reservations, and user profiles.
- Ensured the system supports reservation management, order tracking, and menu updates, contributing to smoother restaurant operations.

Khatm Al-Quran Mobile App

Using Flutter & Dart Language

- Developed a simple app to automate item tracking and distribution for Quran completion (Khatm) within a community group.
- Implemented features to log which Juz' each person received and display what remains.
- Streamlined coordination by replacing manual messaging with an organized, user-friendly interface.

Portfolio

Using React, Vite, Tailwind CSS, Framer Motion, React Scroll, GitHub Pages

- Responsive single-page portfolio with animated sections, smooth scroll navigation, and Framer Motion transitions.
- Includes downloadable resume, lazy-loaded assets, and automated GitHub Pages CI/CD deployment.

SKILLS

Programming Languages: Python, Java, HTML, CSS, JavaScript, TypeScript, C++, C#, Flutter, Dart, R, SQL.

Web & Mobile Development: React, React Native, Angular

Backend & Cloud: Flask, AWS

Tools & Technologies: API Integration, Figma, Canva, Microsoft Office, Jira, Git/GitHub, Postman, Bitbucket

Additional Skills: Scrum Master, Agile/Kanban Methodology, Team Collaboration, UI/UX Design

ACHIEVEMENTS

Joget NextGen Hackathon 2024

No-code / Low-code Hackathon | Oct 2024 – Nov 2024

- 2nd Place: Developed an innovative machinery management application using Joget DX 8 and related technologies.
- Participated in a team of 4, addressing efficient machinery maintenance with a no-code/low-code solution.
- Developed custom UI components using custom HTML within the Joget environment to provide a user-friendly interface for machinery data and maintenance scheduling.

Valedictorian – IGCSE Graduating Class

Wadi Sofia School, 2021

- Awarded Valedictorian for outstanding academic performance, ranking at the top of the graduating class.
- Excelled in key subjects including Mathematics, Chemistry, and Physics, achieving top grades and showing exceptional dedication to academic excellence.